

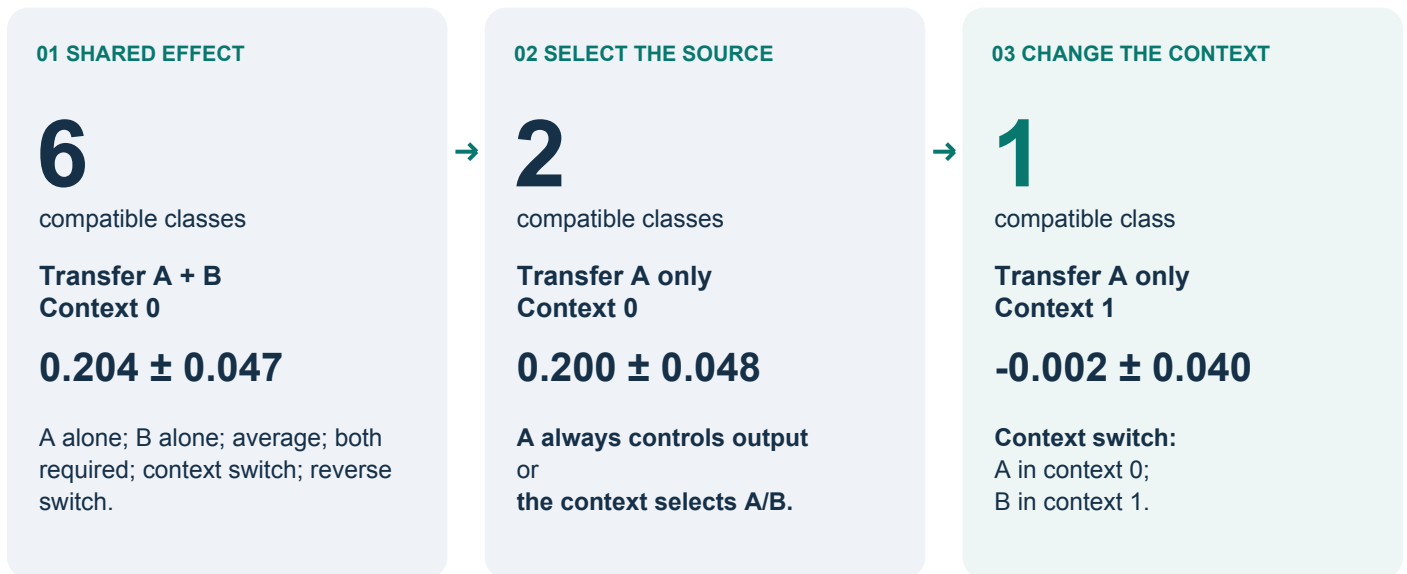
How interventions narrow causal explanations

Felix Borck | A concrete path from one patch effect to fewer compatible explanations

EXISTING DATA • Retrospective replay of one measured synthetic circuit. No new inference.

The question: which internal computation produced the effect?

A and B are two internal channels. All six candidate classes predict the same full-patch output, about 0.199. Selective patches and a context change make their predictions disagree.



Each step retains the earlier measurements. In step 3, A-only predicts about 0.199; the context switch predicts 0. The measured output separates those remaining rivals.

WHAT THE NARROWING ESTABLISHES

One class remains **within the declared candidate set and intervention menu**. Equivalent implementations are treated as one class. This is an illustrative replay, not a validated adaptive selection policy or a uniquely identified mechanism in an LLM.

What already exists in GPT-2

Source comparison: the null-read candidate predicts better in 64/64 fresh pairs (Makelov-derived application). No absolute adequacy threshold was tested. **Later rounds:** simple route and invariance profiles were excluded. These are separate questions, not a single nested identification chain.

The next prospective step

Freeze a next-intervention rule and its error budget; log the surviving set after each measurement; then test the remaining explanation on untouched cases. New rivals or failed controls can reopen the explanation.

Measurement: 8 simulated observations/cell; known independent Gaussian errors. $\pm$ is a menu-wide simultaneous comparison radius ($\alpha=0.005$, 16 cells), plus a numerical allowance; generic scores, not nats. Case fc04b0fee691e4362c18; development_001. Full anchors in both contexts are retained.